

HAT-P-36b

R-1789

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_170330 · created 2026-08-03T00:31:37+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457842.9095 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.117 +/- 0.01
Transit depth (Rp/Rs)^2	1.37 +/- 0.24 [%]
Orbital Inclination (inc)	85.07 +/- 2.31
Airmass coefficient 1 (a1)	0.9887 +/- 0.0097
Airmass coefficient 2 (a2)	0.0097 +/- 0.0062
Scatter in the residuals of the lightcurve fit is	0.98 %
Best Comparison Star	#2 - [84, 424]
Optimal Aperture	3.54
Optimal Annulus	3.84
Transit Duration (day)	0.0896 +/- 0.0062

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.117 $\pm$ 0.01 vs 0.126 (deviation 0.9 σ)
Duration fit vs published	0.0896 d vs 0.0925 d (deviation 0.47 σ)
Mid-time O–C	0.8 min
Depth SNR	11.7
Residual scatter	0.98 %

Light curve

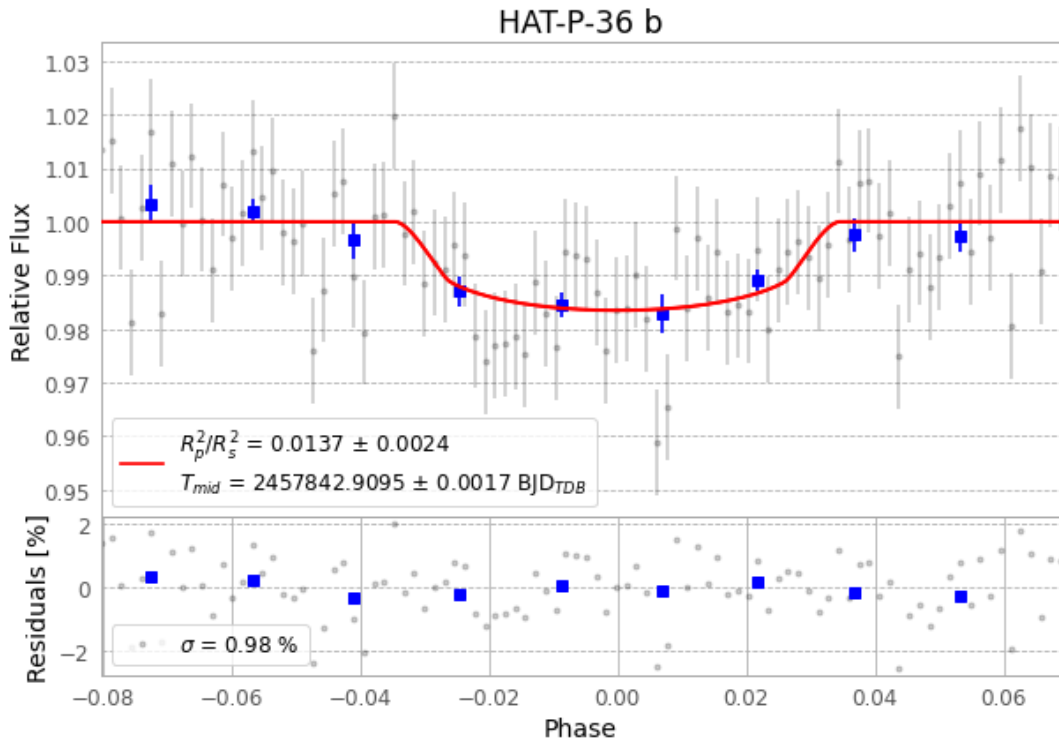


Figure 1 — FinalLightCurve_HAT-P-36 b_2017-03-30.png (SHA-256 6816befd7e3fe117...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [225, 279], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 602627792fe03226...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

96 FITS frames (63 MB), HATP-36170330070912.FITS → HATP-36170330115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-170330.log (661dbe6459f2...), FinalLightCurve_HAT-P-36 b_2017-03-30.png (6816befd7e3f...), FinalLightCurve_HAT-P-36 b_2017-03-30.csv (424385e481cf...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)